

March 26, 2025
File No. 27220090.25

Jennifer Hamby
Barton County Solid Waste
Great Bend, KS 67530

Subject: 2024 Annual NMOC Emission Rate Report
Barton County Sanitary Landfill
Source ID: 0090061

Dear Jennifer:

SCS Engineers has completed the 2024 Annual Non-methane Organic Compound (NMOC) Emission Rate Report for the Barton County Sanitary Landfill (Facility). The Facility is subject to 40 Code of Federal Regulations (CFR) §62, Subpart 000 – *Federal Plan Requirements for Municipal Solid Waste Landfills That Commenced Construction On or Before July 17, 2014 and Have Not Been Modified or Reconstructed Since July 17, 2014*. Pursuant to 40 CFR 62.16724(c), the facility is required to calculate its NMOC emission rate until an active gas collection system is required and installed to comply with 40 CFR 62, Subpart 000.

The annual NMOC emissions from the landfill are calculated using the LandGEM model version 3.02. The model combines site-specific NMOC concentrations from previous Tier 2 testing along with actual in-place waste data to calculate the total NMOC emissions for the year. The 2024 calculations used the following data:

- Actual 2024 waste acceptance: 47,948 Megagrams (Mg)
- Total waste in place: 1,455,460 Mg
- Site-Specific NMOC concentration: 176 parts per million volume as hexane (September 2024 Tier 2)

The resulting NMOC emissions rate for calendar year 2024 was calculated to be 5.628 Mg/year. A copy of the LandGEM model output is attached to this letter. The emission rate remains below the 34 Mg/yr threshold for installing a gas collection and control system (GCCS) pursuant to 40 CFR 62, Subpart 000.



Jennifer Hamby
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If you have any questions regarding this report or calculation, please do not hesitate to contact Marilyn Jones at 913-749-0715.

Sincerely,



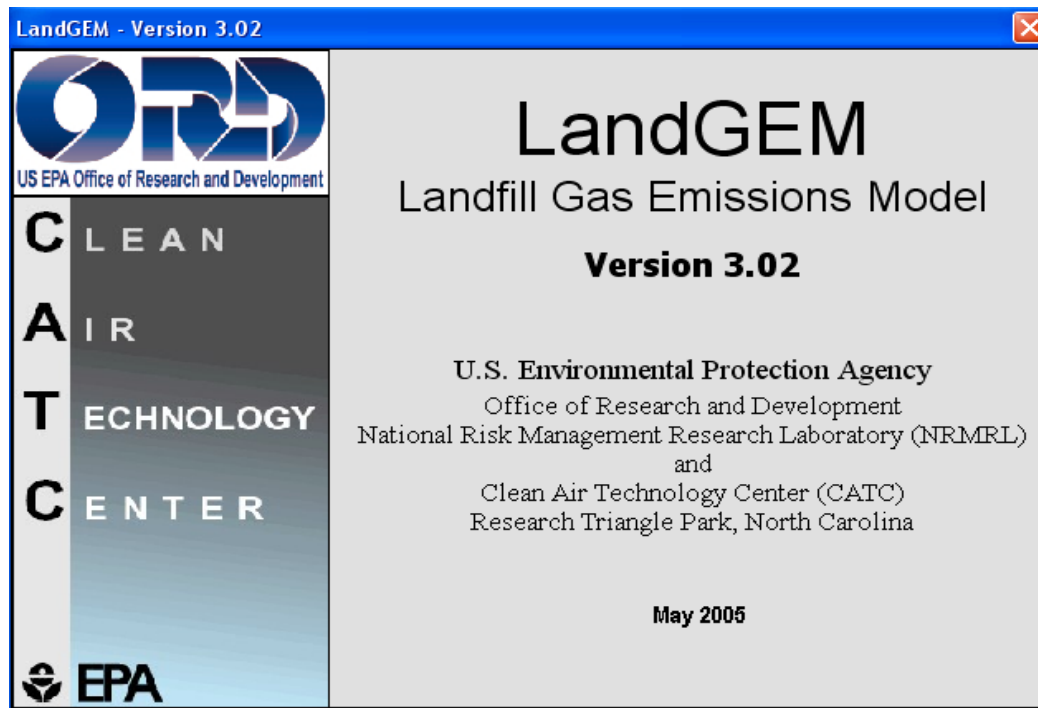
Marilyn Jones
Senior Project Professional
SCS Engineers



Allyson Patterson
Associate Professional
SCS Engineers

Attachment: LandGEM Model Output

Cc'd: EPA Region 7



Summary Report

Landfill Name or Identifier: Barton County

Date: Wednesday, March 26, 2025

Description/Comments:

About LandGEM:

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left(\frac{M_i}{10} \right) e^{-k t_{ij}}$$

Where,

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year
(decimal years, e.g., 3.2 years)

LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of landfilled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate. Defaults for estimating emissions for this type of operation are being developed to include in LandGEM along with defaults for conventional landfills (no leachate or liquid additions) for developing emission inventories and determining CAA applicability. Refer to the Web site identified above for future updates.

Input Review

LANDFILL CHARACTERISTICS

Landfill Open Year	1973
Landfill Closure Year (with 80-year limit)	2052
Actual Closure Year (without limit)	2098
Have Model Calculate Closure Year?	Yes
Waste Design Capacity	4,957,086

The 80-year waste acceptance limit of the model has been exceeded before the Waste Design Capacity was reached. The model will assume the 80th year of waste acceptance as the final year to estimate emissions. See Section 2.6 of the User's Manual.

megagrams

MODEL PARAMETERS

Methane Generation Rate, k	0.050	year ⁻¹
Potential Methane Generation Capacity, L ₀	170	m ³ /Mg
NMOC Concentration	176	ppmv as hexane
Methane Content	50	% by volume

GASES / POLLUTANTS SELECTED

Gas / Pollutant #1:	Total landfill gas
Gas / Pollutant #2:	Methane
Gas / Pollutant #3:	Carbon dioxide
Gas / Pollutant #4:	NMOC

WASTE ACCEPTANCE RATES

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
1973	21,207	23,328	0	0
1974	23,173	25,490	21,207	23,328
1975	24,296	26,726	44,380	48,818
1976	24,036	26,440	68,676	75,544
1977	19,342	21,276	92,712	101,983
1978	23,878	26,266	112,054	123,259
1979	23,290	25,619	135,932	149,525
1980	22,962	25,258	159,222	175,144
1981	32,840	36,124	182,184	200,402
1982	26,462	29,108	215,024	236,526
1983	24,371	26,808	241,486	265,635
1984	23,790	26,169	265,857	292,443
1985	24,975	27,473	289,647	318,612
1986	25,667	28,234	314,622	346,084
1987	23,577	25,935	340,289	374,318
1988	23,996	26,396	363,866	400,253
1989	24,191	26,610	387,862	426,648
1990	24,436	26,880	412,053	453,258
1991	25,718	28,290	436,489	480,138
1992	24,385	26,824	462,207	508,428
1993	28,583	31,441	486,592	535,251
1994	29,599	32,559	515,175	566,693
1995	29,298	32,228	544,774	599,251
1996	31,253	34,378	574,072	631,479
1997	40,919	45,011	605,325	665,858
1998	35,972	39,569	646,244	710,868
1999	37,645	41,410	682,216	750,438
2000	47,533	52,286	719,861	791,847
2001	33,831	37,214	767,394	844,133
2002	22,829	25,112	801,225	881,348
2003	21,717	23,889	824,054	906,459
2004	26,357	28,993	845,771	930,348
2005	26,080	28,688	872,128	959,341
2006	24,395	26,835	898,208	988,029
2007	26,111	28,722	922,603	1,014,863
2008	24,910	27,401	948,714	1,043,585
2009	32,650	35,915	973,624	1,070,986
2010	25,989	28,587	1,006,274	1,106,901
2011	35,131	38,645	1,032,263	1,135,489
2012	24,755	27,230	1,067,394	1,174,133

WASTE ACCEPTANCE RATES (Continued)

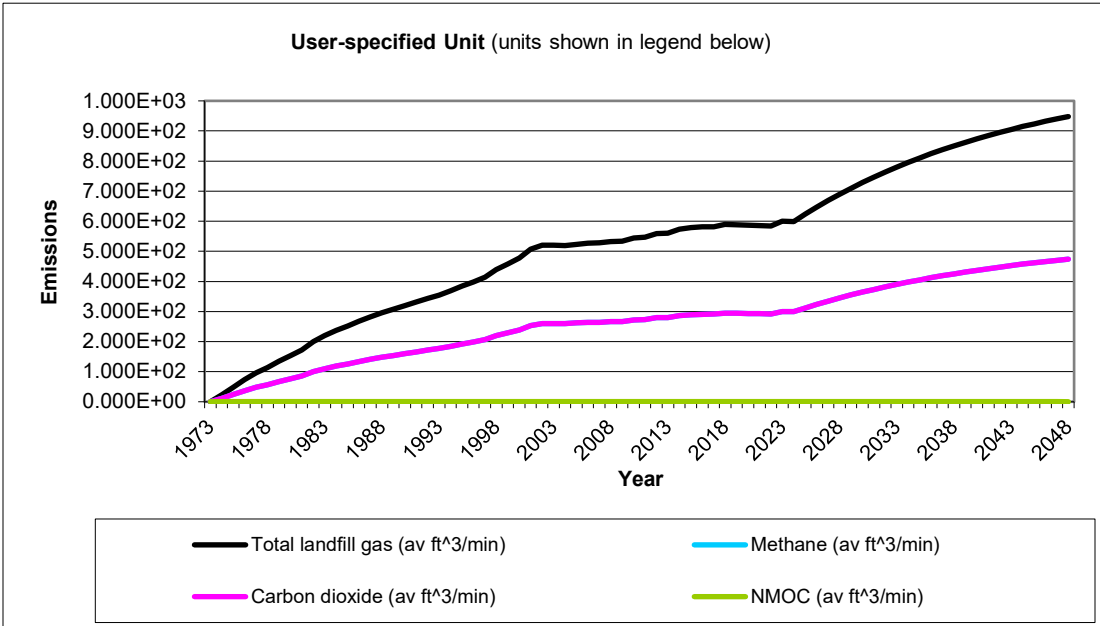
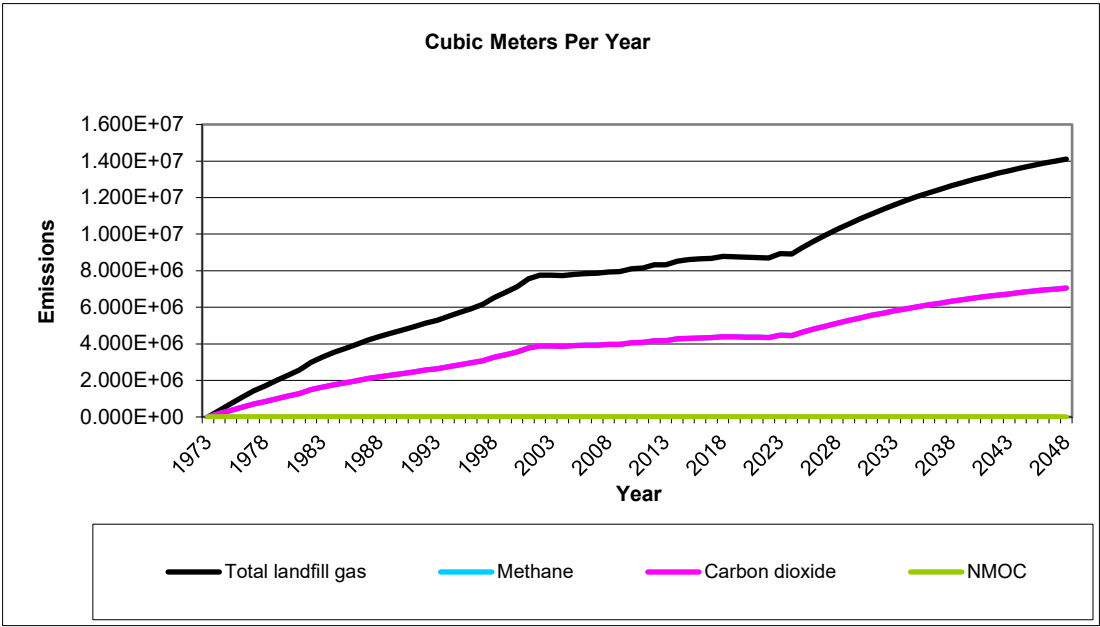
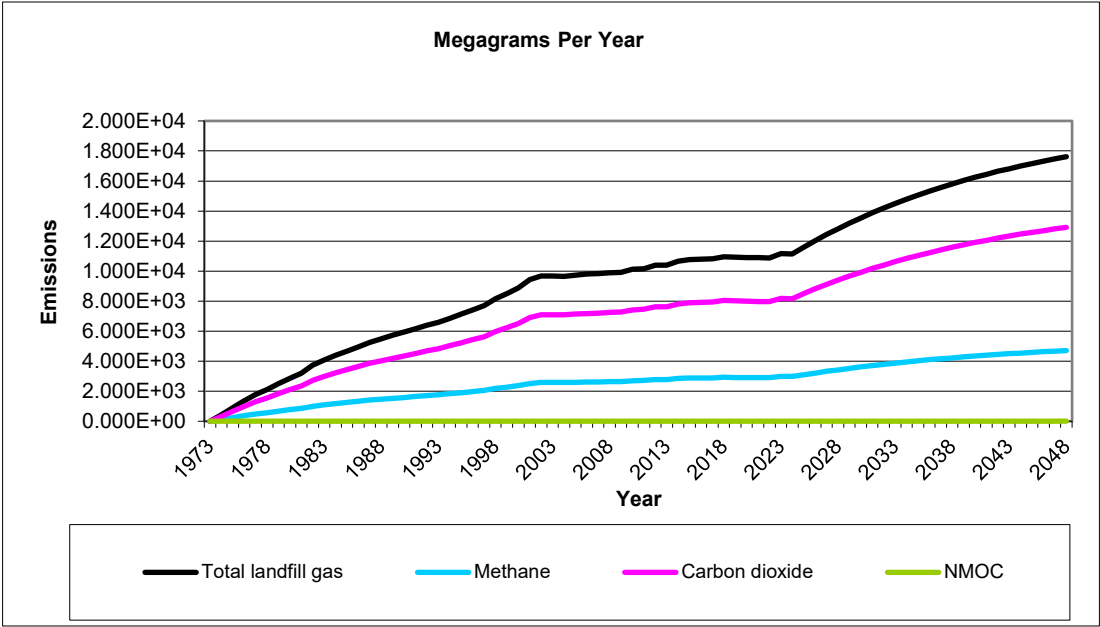
Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2013	36,476	40,124	1,092,149	1,201,364
2014	30,173	33,190	1,128,625	1,241,488
2015	26,967	29,664	1,158,798	1,274,678
2016	26,516	29,168	1,185,765	1,304,341
2017	32,331	35,564	1,212,281	1,333,510
2018	24,136	26,550	1,244,613	1,369,074
2019	24,306	26,736	1,268,749	1,395,624
2020	24,970	27,467	1,293,054	1,422,360
2021	24,373	26,810	1,318,024	1,449,827
2022	39,978	43,976	1,342,397	1,476,637
2023	25,137	27,651	1,382,375	1,520,612
2024	47,948	52,743	1,407,512	1,548,263
2025	47,948	52,743	1,455,460	1,601,006
2026	47,948	52,743	1,503,408	1,653,749
2027	47,948	52,743	1,551,357	1,706,492
2028	47,948	52,743	1,599,305	1,759,235
2029	47,948	52,743	1,647,253	1,811,978
2030	47,948	52,743	1,695,201	1,864,721
2031	47,948	52,743	1,743,149	1,917,464
2032	47,948	52,743	1,791,098	1,970,207
2033	47,948	52,743	1,839,046	2,022,951
2034	47,948	52,743	1,886,994	2,075,694
2035	47,948	52,743	1,934,942	2,128,437
2036	47,948	52,743	1,982,891	2,181,180
2037	47,948	52,743	2,030,839	2,233,923
2038	47,948	52,743	2,078,787	2,286,666
2039	47,948	52,743	2,126,735	2,339,409
2040	47,948	52,743	2,174,683	2,392,152
2041	47,948	52,743	2,222,632	2,444,895
2042	47,948	52,743	2,270,580	2,497,638
2043	47,948	52,743	2,318,528	2,550,381
2044	47,948	52,743	2,366,476	2,603,124
2045	47,948	52,743	2,414,424	2,655,867
2046	47,948	52,743	2,462,373	2,708,610
2047	47,948	52,743	2,510,321	2,761,353
2048	47,948	52,743	2,558,269	2,814,096
2049	47,948	52,743	2,606,217	2,866,839
2050	47,948	52,743	2,654,165	2,919,582
2051	47,948	52,743	2,702,114	2,972,325
2052	47,948	52,743	2,750,062	3,025,068

Pollutant Parameters

Gas / Pollutant Default Parameters:				User-specified Pollutant Parameters:	
	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Gases	Total landfill gas	4,000	0.00		
	Methane		16.04		
	Carbon dioxide		44.01		
	NMOC		86.18		
Pollutants	1,1,1-Trichloroethane (methyl chloroform) - HAP	0.48	133.41		
	1,1,2,2- Tetrachloroethane - HAP/VOC	1.1	167.85		
	1,1-Dichloroethane (ethylidene dichloride) - HAP/VOC	2.4	98.97		
	1,1-Dichloroethene (vinylidene chloride) - HAP/VOC	0.20	96.94		
	1,2-Dichloroethane (ethylene dichloride) - HAP/VOC	0.41	98.96		
	1,2-Dichloropropane (propylene dichloride) - HAP/VOC	0.18	112.99		
	2-Propanol (isopropyl alcohol) - VOC	50	60.11		
	Acetone	7.0	58.08		
	Acrylonitrile - HAP/VOC	6.3	53.06		
	Benzene - No or Unknown Co-disposal - HAP/VOC	1.9	78.11		
	Benzene - Co-disposal - HAP/VOC	11	78.11		
	Bromodichloromethane - VOC	3.1	163.83		
	Butane - VOC	5.0	58.12		
	Carbon disulfide - HAP/VOC	0.58	76.13		
	Carbon monoxide	140	28.01		
	Carbon tetrachloride - HAP/VOC	4.0E-03	153.84		
	Carbonyl sulfide - HAP/VOC	0.49	60.07		
	Chlorobenzene - HAP/VOC	0.25	112.56		
	Chlorodifluoromethane	1.3	86.47		
	Chloroethane (ethyl chloride) - HAP/VOC	1.3	64.52		
	Chloroform - HAP/VOC	0.03	119.39		
	Chloromethane - VOC	1.2	50.49		
	Dichlorobenzene - (HAP for para isomer/VOC)	0.21	147		
	Dichlorodifluoromethane	16	120.91		
	Dichlorofluoromethane - VOC	2.6	102.92		
	Dichloromethane (methylene chloride) - HAP	14	84.94		
	Dimethyl sulfide (methyl sulfide) - VOC	7.8	62.13		
	Ethane	890	30.07		
	Ethanol - VOC	27	46.08		

[illegible]

Graphs



Results

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1973	0	0	0	0	0	0
1974	4.403E+02	3.525E+05	2.369E+01	1.176E+02	1.763E+05	1.184E+01
1975	8.998E+02	7.206E+05	4.841E+01	2.404E+02	3.603E+05	2.421E+01
1976	1.360E+03	1.089E+06	7.319E+01	3.634E+02	5.446E+05	3.659E+01
1977	1.793E+03	1.436E+06	9.647E+01	4.789E+02	7.179E+05	4.823E+01
1978	2.107E+03	1.687E+06	1.134E+02	5.628E+02	8.436E+05	5.668E+01
1979	2.500E+03	2.002E+06	1.345E+02	6.678E+02	1.001E+06	6.725E+01
1980	2.862E+03	2.291E+06	1.540E+02	7.644E+02	1.146E+06	7.698E+01
1981	3.199E+03	2.561E+06	1.721E+02	8.544E+02	1.281E+06	8.605E+01
1982	3.724E+03	2.982E+06	2.004E+02	9.948E+02	1.491E+06	1.002E+02
1983	4.092E+03	3.277E+06	2.202E+02	1.093E+03	1.638E+06	1.101E+02
1984	4.399E+03	3.522E+06	2.367E+02	1.175E+03	1.761E+06	1.183E+02
1985	4.678E+03	3.746E+06	2.517E+02	1.250E+03	1.873E+06	1.258E+02
1986	4.968E+03	3.978E+06	2.673E+02	1.327E+03	1.989E+06	1.337E+02
1987	5.259E+03	4.211E+06	2.829E+02	1.405E+03	2.105E+06	1.415E+02
1988	5.492E+03	4.398E+06	2.955E+02	1.467E+03	2.199E+06	1.477E+02
1989	5.722E+03	4.582E+06	3.079E+02	1.528E+03	2.291E+06	1.539E+02
1990	5.945E+03	4.761E+06	3.199E+02	1.588E+03	2.380E+06	1.599E+02
1991	6.163E+03	4.935E+06	3.316E+02	1.646E+03	2.467E+06	1.658E+02
1992	6.396E+03	5.122E+06	3.441E+02	1.708E+03	2.561E+06	1.721E+02
1993	6.590E+03	5.277E+06	3.546E+02	1.760E+03	2.639E+06	1.773E+02
1994	6.862E+03	5.495E+06	3.692E+02	1.833E+03	2.747E+06	1.846E+02
1995	7.142E+03	5.719E+06	3.843E+02	1.908E+03	2.859E+06	1.921E+02
1996	7.402E+03	5.927E+06	3.982E+02	1.977E+03	2.964E+06	1.991E+02
1997	7.690E+03	6.158E+06	4.137E+02	2.054E+03	3.079E+06	2.069E+02
1998	8.164E+03	6.537E+06	4.392E+02	2.181E+03	3.269E+06	2.196E+02
1999	8.513E+03	6.817E+06	4.580E+02	2.274E+03	3.408E+06	2.290E+02
2000	8.879E+03	7.110E+06	4.777E+02	2.372E+03	3.555E+06	2.389E+02
2001	9.433E+03	7.553E+06	5.075E+02	2.520E+03	3.777E+06	2.538E+02
2002	9.675E+03	7.747E+06	5.205E+02	2.584E+03	3.874E+06	2.603E+02
2003	9.677E+03	7.749E+06	5.207E+02	2.585E+03	3.874E+06	2.603E+02
2004	9.656E+03	7.732E+06	5.195E+02	2.579E+03	3.866E+06	2.598E+02
2005	9.732E+03	7.793E+06	5.236E+02	2.600E+03	3.897E+06	2.618E+02
2006	9.799E+03	7.847E+06	5.272E+02	2.617E+03	3.923E+06	2.636E+02
2007	9.828E+03	7.869E+06	5.287E+02	2.625E+03	3.935E+06	2.644E+02
2008	9.890E+03	7.920E+06	5.321E+02	2.642E+03	3.960E+06	2.661E+02
2009	9.925E+03	7.948E+06	5.340E+02	2.651E+03	3.974E+06	2.670E+02
2010	1.012E+04	8.103E+06	5.444E+02	2.703E+03	4.051E+06	2.722E+02
2011	1.016E+04	8.140E+06	5.469E+02	2.715E+03	4.070E+06	2.734E+02
2012	1.040E+04	8.327E+06	5.595E+02	2.778E+03	4.163E+06	2.797E+02
2013	1.041E+04	8.332E+06	5.598E+02	2.779E+03	4.166E+06	2.799E+02
2014	1.065E+04	8.532E+06	5.733E+02	2.846E+03	4.266E+06	2.866E+02
2015	1.076E+04	8.617E+06	5.790E+02	2.875E+03	4.309E+06	2.895E+02
2016	1.080E+04	8.645E+06	5.809E+02	2.884E+03	4.323E+06	2.904E+02
2017	1.082E+04	8.665E+06	5.822E+02	2.890E+03	4.332E+06	2.911E+02
2018	1.096E+04	8.780E+06	5.899E+02	2.929E+03	4.390E+06	2.949E+02
2019	1.093E+04	8.753E+06	5.881E+02	2.920E+03	4.376E+06	2.940E+02
2020	1.090E+04	8.730E+06	5.865E+02	2.912E+03	4.365E+06	2.933E+02
2021	1.089E+04	8.719E+06	5.858E+02	2.908E+03	4.360E+06	2.929E+02
2022	1.086E+04	8.699E+06	5.845E+02	2.902E+03	4.349E+06	2.922E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2023	1.116E+04	8.939E+06	6.006E+02	2.982E+03	4.470E+06	3.003E+02
2024	1.114E+04	8.921E+06	5.994E+02	2.976E+03	4.461E+06	2.997E+02
2025	1.159E+04	9.283E+06	6.237E+02	3.097E+03	4.642E+06	3.119E+02
2026	1.202E+04	9.627E+06	6.469E+02	3.211E+03	4.814E+06	3.234E+02
2027	1.243E+04	9.955E+06	6.689E+02	3.321E+03	4.978E+06	3.344E+02
2028	1.282E+04	1.027E+07	6.898E+02	3.425E+03	5.133E+06	3.449E+02
2029	1.319E+04	1.056E+07	7.097E+02	3.524E+03	5.281E+06	3.549E+02
2030	1.354E+04	1.084E+07	7.287E+02	3.618E+03	5.422E+06	3.643E+02
2031	1.388E+04	1.111E+07	7.467E+02	3.707E+03	5.556E+06	3.733E+02
2032	1.420E+04	1.137E+07	7.638E+02	3.792E+03	5.684E+06	3.819E+02
2033	1.450E+04	1.161E+07	7.801E+02	3.873E+03	5.805E+06	3.901E+02
2034	1.479E+04	1.184E+07	7.956E+02	3.950E+03	5.921E+06	3.978E+02
2035	1.506E+04	1.206E+07	8.104E+02	4.023E+03	6.031E+06	4.052E+02
2036	1.532E+04	1.227E+07	8.244E+02	4.093E+03	6.135E+06	4.122E+02
2037	1.557E+04	1.247E+07	8.378E+02	4.159E+03	6.234E+06	4.189E+02
2038	1.581E+04	1.266E+07	8.505E+02	4.222E+03	6.329E+06	4.252E+02
2039	1.603E+04	1.284E+07	8.625E+02	4.282E+03	6.419E+06	4.313E+02
2040	1.625E+04	1.301E+07	8.740E+02	4.339E+03	6.504E+06	4.370E+02
2041	1.645E+04	1.317E+07	8.850E+02	4.393E+03	6.585E+06	4.425E+02
2042	1.664E+04	1.333E+07	8.953E+02	4.445E+03	6.663E+06	4.477E+02
2043	1.683E+04	1.347E+07	9.052E+02	4.494E+03	6.736E+06	4.526E+02
2044	1.700E+04	1.361E+07	9.146E+02	4.541E+03	6.806E+06	4.573E+02
2045	1.717E+04	1.375E+07	9.236E+02	4.585E+03	6.873E+06	4.618E+02
2046	1.732E+04	1.387E+07	9.321E+02	4.628E+03	6.936E+06	4.660E+02
2047	1.747E+04	1.399E+07	9.402E+02	4.668E+03	6.997E+06	4.701E+02
2048	1.762E+04	1.411E+07	9.479E+02	4.706E+03	7.054E+06	4.739E+02
2049	1.775E+04	1.422E+07	9.552E+02	4.742E+03	7.108E+06	4.776E+02
2050	1.788E+04	1.432E+07	9.622E+02	4.777E+03	7.160E+06	4.811E+02
2051	1.801E+04	1.442E+07	9.688E+02	4.810E+03	7.210E+06	4.844E+02
2052	1.812E+04	1.451E+07	9.751E+02	4.841E+03	7.256E+06	4.876E+02
2053	1.824E+04	1.460E+07	9.811E+02	4.871E+03	7.301E+06	4.906E+02
2054	1.735E+04	1.389E+07	9.333E+02	4.633E+03	6.945E+06	4.666E+02
2055	1.650E+04	1.321E+07	8.878E+02	4.407E+03	6.606E+06	4.439E+02
2056	1.570E+04	1.257E+07	8.445E+02	4.192E+03	6.284E+06	4.222E+02
2057	1.493E+04	1.196E+07	8.033E+02	3.988E+03	5.978E+06	4.016E+02
2058	1.420E+04	1.137E+07	7.641E+02	3.793E+03	5.686E+06	3.820E+02
2059	1.351E+04	1.082E+07	7.268E+02	3.608E+03	5.409E+06	3.634E+02
2060	1.285E+04	1.029E+07	6.914E+02	3.432E+03	5.145E+06	3.457E+02
2061	1.222E+04	9.788E+06	6.577E+02	3.265E+03	4.894E+06	3.288E+02
2062	1.163E+04	9.311E+06	6.256E+02	3.106E+03	4.655E+06	3.128E+02
2063	1.106E+04	8.857E+06	5.951E+02	2.954E+03	4.428E+06	2.975E+02
2064	1.052E+04	8.425E+06	5.661E+02	2.810E+03	4.212E+06	2.830E+02
2065	1.001E+04	8.014E+06	5.385E+02	2.673E+03	4.007E+06	2.692E+02
2066	9.520E+03	7.623E+06	5.122E+02	2.543E+03	3.812E+06	2.561E+02
2067	9.056E+03	7.251E+06	4.872E+02	2.419E+03	3.626E+06	2.436E+02
2068	8.614E+03	6.898E+06	4.634E+02	2.301E+03	3.449E+06	2.317E+02
2069	8.194E+03	6.561E+06	4.408E+02	2.189E+03	3.281E+06	2.204E+02
2070	7.794E+03	6.241E+06	4.193E+02	2.082E+03	3.121E+06	2.097E+02
2071	7.414E+03	5.937E+06	3.989E+02	1.980E+03	2.968E+06	1.994E+02
2072	7.052E+03	5.647E+06	3.794E+02	1.884E+03	2.824E+06	1.897E+02
2073	6.708E+03	5.372E+06	3.609E+02	1.792E+03	2.686E+06	1.805E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2074	6.381E+03	5.110E+06	3.433E+02	1.705E+03	2.555E+06	1.717E+02
2075	6.070E+03	4.861E+06	3.266E+02	1.621E+03	2.430E+06	1.633E+02
2076	5.774E+03	4.624E+06	3.107E+02	1.542E+03	2.312E+06	1.553E+02
2077	5.492E+03	4.398E+06	2.955E+02	1.467E+03	2.199E+06	1.478E+02
2078	5.225E+03	4.184E+06	2.811E+02	1.396E+03	2.092E+06	1.405E+02
2079	4.970E+03	3.980E+06	2.674E+02	1.327E+03	1.990E+06	1.337E+02
2080	4.727E+03	3.785E+06	2.543E+02	1.263E+03	1.893E+06	1.272E+02
2081	4.497E+03	3.601E+06	2.419E+02	1.201E+03	1.800E+06	1.210E+02
2082	4.278E+03	3.425E+06	2.301E+02	1.143E+03	1.713E+06	1.151E+02
2083	4.069E+03	3.258E+06	2.189E+02	1.087E+03	1.629E+06	1.095E+02
2084	3.870E+03	3.099E+06	2.082E+02	1.034E+03	1.550E+06	1.041E+02
2085	3.682E+03	2.948E+06	1.981E+02	9.834E+02	1.474E+06	9.904E+01
2086	3.502E+03	2.804E+06	1.884E+02	9.355E+02	1.402E+06	9.421E+01
2087	3.331E+03	2.668E+06	1.792E+02	8.898E+02	1.334E+06	8.962E+01
2088	3.169E+03	2.537E+06	1.705E+02	8.464E+02	1.269E+06	8.525E+01
2089	3.014E+03	2.414E+06	1.622E+02	8.052E+02	1.207E+06	8.109E+01
2090	2.867E+03	2.296E+06	1.543E+02	7.659E+02	1.148E+06	7.713E+01
2091	2.727E+03	2.184E+06	1.467E+02	7.285E+02	1.092E+06	7.337E+01
2092	2.594E+03	2.078E+06	1.396E+02	6.930E+02	1.039E+06	6.979E+01
2093	2.468E+03	1.976E+06	1.328E+02	6.592E+02	9.881E+05	6.639E+01
2094	2.348E+03	1.880E+06	1.263E+02	6.271E+02	9.399E+05	6.315E+01
2095	2.233E+03	1.788E+06	1.201E+02	5.965E+02	8.941E+05	6.007E+01
2096	2.124E+03	1.701E+06	1.143E+02	5.674E+02	8.505E+05	5.714E+01
2097	2.021E+03	1.618E+06	1.087E+02	5.397E+02	8.090E+05	5.436E+01
2098	1.922E+03	1.539E+06	1.034E+02	5.134E+02	7.695E+05	5.170E+01
2099	1.828E+03	1.464E+06	9.837E+01	4.884E+02	7.320E+05	4.918E+01
2100	1.739E+03	1.393E+06	9.357E+01	4.645E+02	6.963E+05	4.678E+01
2101	1.654E+03	1.325E+06	8.901E+01	4.419E+02	6.623E+05	4.450E+01
2102	1.574E+03	1.260E+06	8.466E+01	4.203E+02	6.300E+05	4.233E+01
2103	1.497E+03	1.199E+06	8.054E+01	3.998E+02	5.993E+05	4.027E+01
2104	1.424E+03	1.140E+06	7.661E+01	3.803E+02	5.701E+05	3.830E+01
2105	1.354E+03	1.085E+06	7.287E+01	3.618E+02	5.423E+05	3.644E+01
2106	1.288E+03	1.032E+06	6.932E+01	3.441E+02	5.158E+05	3.466E+01
2107	1.226E+03	9.813E+05	6.594E+01	3.274E+02	4.907E+05	3.297E+01
2108	1.166E+03	9.335E+05	6.272E+01	3.114E+02	4.667E+05	3.136E+01
2109	1.109E+03	8.880E+05	5.966E+01	2.962E+02	4.440E+05	2.983E+01
2110	1.055E+03	8.447E+05	5.675E+01	2.818E+02	4.223E+05	2.838E+01
2111	1.003E+03	8.035E+05	5.398E+01	2.680E+02	4.017E+05	2.699E+01
2112	9.544E+02	7.643E+05	5.135E+01	2.549E+02	3.821E+05	2.568E+01
2113	9.079E+02	7.270E+05	4.885E+01	2.425E+02	3.635E+05	2.442E+01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1973	0	0	0	0	0	0
1974	3.227E+02	1.763E+05	1.184E+01	2.224E-01	6.205E+01	4.169E-03
1975	6.595E+02	3.603E+05	2.421E+01	4.546E-01	1.268E+02	8.521E-03
1976	9.970E+02	5.446E+05	3.659E+01	6.872E-01	1.917E+02	1.288E-02
1977	1.314E+03	7.179E+05	4.823E+01	9.058E-01	2.527E+02	1.698E-02
1978	1.544E+03	8.436E+05	5.668E+01	1.064E+00	2.970E+02	1.995E-02
1979	1.832E+03	1.001E+06	6.725E+01	1.263E+00	3.523E+02	2.367E-02
1980	2.097E+03	1.146E+06	7.698E+01	1.446E+00	4.033E+02	2.710E-02
1981	2.344E+03	1.281E+06	8.605E+01	1.616E+00	4.508E+02	3.029E-02
1982	2.730E+03	1.491E+06	1.002E+02	1.881E+00	5.249E+02	3.527E-02
1983	2.999E+03	1.638E+06	1.101E+02	2.067E+00	5.767E+02	3.875E-02
1984	3.224E+03	1.761E+06	1.183E+02	2.222E+00	6.199E+02	4.165E-02
1985	3.428E+03	1.873E+06	1.258E+02	2.363E+00	6.593E+02	4.430E-02
1986	3.641E+03	1.989E+06	1.337E+02	2.510E+00	7.002E+02	4.705E-02
1987	3.854E+03	2.105E+06	1.415E+02	2.657E+00	7.411E+02	4.980E-02
1988	4.025E+03	2.199E+06	1.477E+02	2.774E+00	7.740E+02	5.200E-02
1989	4.194E+03	2.291E+06	1.539E+02	2.891E+00	8.064E+02	5.418E-02
1990	4.357E+03	2.380E+06	1.599E+02	3.003E+00	8.379E+02	5.630E-02
1991	4.516E+03	2.467E+06	1.658E+02	3.113E+00	8.685E+02	5.835E-02
1992	4.687E+03	2.561E+06	1.721E+02	3.231E+00	9.014E+02	6.056E-02
1993	4.830E+03	2.639E+06	1.773E+02	3.329E+00	9.288E+02	6.240E-02
1994	5.029E+03	2.747E+06	1.846E+02	3.467E+00	9.671E+02	6.498E-02
1995	5.234E+03	2.859E+06	1.921E+02	3.608E+00	1.007E+03	6.763E-02
1996	5.425E+03	2.964E+06	1.991E+02	3.739E+00	1.043E+03	7.009E-02
1997	5.636E+03	3.079E+06	2.069E+02	3.885E+00	1.084E+03	7.282E-02
1998	5.983E+03	3.269E+06	2.196E+02	4.124E+00	1.151E+03	7.731E-02
1999	6.239E+03	3.408E+06	2.290E+02	4.300E+00	1.200E+03	8.061E-02
2000	6.507E+03	3.555E+06	2.389E+02	4.485E+00	1.251E+03	8.408E-02
2001	6.913E+03	3.777E+06	2.538E+02	4.765E+00	1.329E+03	8.932E-02
2002	7.091E+03	3.874E+06	2.603E+02	4.888E+00	1.364E+03	9.162E-02
2003	7.092E+03	3.874E+06	2.603E+02	4.889E+00	1.364E+03	9.164E-02
2004	7.077E+03	3.866E+06	2.598E+02	4.878E+00	1.361E+03	9.144E-02
2005	7.133E+03	3.897E+06	2.618E+02	4.916E+00	1.372E+03	9.216E-02
2006	7.182E+03	3.923E+06	2.636E+02	4.950E+00	1.381E+03	9.279E-02
2007	7.203E+03	3.935E+06	2.644E+02	4.965E+00	1.385E+03	9.306E-02
2008	7.248E+03	3.960E+06	2.661E+02	4.996E+00	1.394E+03	9.365E-02
2009	7.274E+03	3.974E+06	2.670E+02	5.014E+00	1.399E+03	9.398E-02
2010	7.416E+03	4.051E+06	2.722E+02	5.112E+00	1.426E+03	9.582E-02
2011	7.450E+03	4.070E+06	2.734E+02	5.135E+00	1.433E+03	9.625E-02
2012	7.621E+03	4.163E+06	2.797E+02	5.253E+00	1.465E+03	9.847E-02
2013	7.626E+03	4.166E+06	2.799E+02	5.256E+00	1.466E+03	9.853E-02
2014	7.809E+03	4.266E+06	2.866E+02	5.383E+00	1.502E+03	1.009E-01
2015	7.887E+03	4.309E+06	2.895E+02	5.436E+00	1.517E+03	1.019E-01
2016	7.913E+03	4.323E+06	2.904E+02	5.454E+00	1.522E+03	1.022E-01
2017	7.930E+03	4.332E+06	2.911E+02	5.466E+00	1.525E+03	1.025E-01
2018	8.035E+03	4.390E+06	2.949E+02	5.539E+00	1.545E+03	1.038E-01
2019	8.011E+03	4.376E+06	2.940E+02	5.522E+00	1.540E+03	1.035E-01
2020	7.990E+03	4.365E+06	2.933E+02	5.507E+00	1.536E+03	1.032E-01
2021	7.980E+03	4.360E+06	2.929E+02	5.501E+00	1.535E+03	1.031E-01
2022	7.962E+03	4.349E+06	2.922E+02	5.488E+00	1.531E+03	1.029E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2023	8.182E+03	4.470E+06	3.003E+02	5.640E+00	1.573E+03	1.057E-01
2024	8.165E+03	4.461E+06	2.997E+02	5.628E+00	1.570E+03	1.055E-01
2025	8.496E+03	4.642E+06	3.119E+02	5.856E+00	1.634E+03	1.098E-01
2026	8.812E+03	4.814E+06	3.234E+02	6.074E+00	1.694E+03	1.138E-01
2027	9.111E+03	4.978E+06	3.344E+02	6.280E+00	1.752E+03	1.177E-01
2028	9.396E+03	5.133E+06	3.449E+02	6.477E+00	1.807E+03	1.214E-01
2029	9.668E+03	5.281E+06	3.549E+02	6.664E+00	1.859E+03	1.249E-01
2030	9.926E+03	5.422E+06	3.643E+02	6.842E+00	1.909E+03	1.282E-01
2031	1.017E+04	5.556E+06	3.733E+02	7.011E+00	1.956E+03	1.314E-01
2032	1.040E+04	5.684E+06	3.819E+02	7.172E+00	2.001E+03	1.344E-01
2033	1.063E+04	5.805E+06	3.901E+02	7.325E+00	2.043E+03	1.373E-01
2034	1.084E+04	5.921E+06	3.978E+02	7.470E+00	2.084E+03	1.400E-01
2035	1.104E+04	6.031E+06	4.052E+02	7.609E+00	2.123E+03	1.426E-01
2036	1.123E+04	6.135E+06	4.122E+02	7.741E+00	2.160E+03	1.451E-01
2037	1.141E+04	6.234E+06	4.189E+02	7.866E+00	2.194E+03	1.474E-01
2038	1.158E+04	6.329E+06	4.252E+02	7.985E+00	2.228E+03	1.497E-01
2039	1.175E+04	6.419E+06	4.313E+02	8.099E+00	2.259E+03	1.518E-01
2040	1.191E+04	6.504E+06	4.370E+02	8.206E+00	2.289E+03	1.538E-01
2041	1.205E+04	6.585E+06	4.425E+02	8.309E+00	2.318E+03	1.558E-01
2042	1.220E+04	6.663E+06	4.477E+02	8.407E+00	2.345E+03	1.576E-01
2043	1.233E+04	6.736E+06	4.526E+02	8.500E+00	2.371E+03	1.593E-01
2044	1.246E+04	6.806E+06	4.573E+02	8.588E+00	2.396E+03	1.610E-01
2045	1.258E+04	6.873E+06	4.618E+02	8.672E+00	2.419E+03	1.626E-01
2046	1.270E+04	6.936E+06	4.660E+02	8.752E+00	2.442E+03	1.640E-01
2047	1.281E+04	6.997E+06	4.701E+02	8.828E+00	2.463E+03	1.655E-01
2048	1.291E+04	7.054E+06	4.739E+02	8.900E+00	2.483E+03	1.668E-01
2049	1.301E+04	7.108E+06	4.776E+02	8.969E+00	2.502E+03	1.681E-01
2050	1.311E+04	7.160E+06	4.811E+02	9.034E+00	2.520E+03	1.693E-01
2051	1.320E+04	7.210E+06	4.844E+02	9.097E+00	2.538E+03	1.705E-01
2052	1.328E+04	7.256E+06	4.876E+02	9.156E+00	2.554E+03	1.716E-01
2053	1.336E+04	7.301E+06	4.906E+02	9.212E+00	2.570E+03	1.727E-01
2054	1.271E+04	6.945E+06	4.666E+02	8.763E+00	2.445E+03	1.643E-01
2055	1.209E+04	6.606E+06	4.439E+02	8.335E+00	2.325E+03	1.562E-01
2056	1.150E+04	6.284E+06	4.222E+02	7.929E+00	2.212E+03	1.486E-01
2057	1.094E+04	5.978E+06	4.016E+02	7.542E+00	2.104E+03	1.414E-01
2058	1.041E+04	5.686E+06	3.820E+02	7.174E+00	2.002E+03	1.345E-01
2059	9.901E+03	5.409E+06	3.634E+02	6.824E+00	1.904E+03	1.279E-01
2060	9.418E+03	5.145E+06	3.457E+02	6.492E+00	1.811E+03	1.217E-01
2061	8.959E+03	4.894E+06	3.288E+02	6.175E+00	1.723E+03	1.157E-01
2062	8.522E+03	4.655E+06	3.128E+02	5.874E+00	1.639E+03	1.101E-01
2063	8.106E+03	4.428E+06	2.975E+02	5.587E+00	1.559E+03	1.047E-01
2064	7.711E+03	4.212E+06	2.830E+02	5.315E+00	1.483E+03	9.963E-02
2065	7.335E+03	4.007E+06	2.692E+02	5.056E+00	1.410E+03	9.477E-02
2066	6.977E+03	3.812E+06	2.561E+02	4.809E+00	1.342E+03	9.015E-02
2067	6.637E+03	3.626E+06	2.436E+02	4.575E+00	1.276E+03	8.575E-02
2068	6.313E+03	3.449E+06	2.317E+02	4.351E+00	1.214E+03	8.157E-02
2069	6.005E+03	3.281E+06	2.204E+02	4.139E+00	1.155E+03	7.759E-02
2070	5.712E+03	3.121E+06	2.097E+02	3.937E+00	1.098E+03	7.380E-02
2071	5.434E+03	2.968E+06	1.994E+02	3.745E+00	1.045E+03	7.021E-02
2072	5.169E+03	2.824E+06	1.897E+02	3.563E+00	9.939E+02	6.678E-02
2073	4.917E+03	2.686E+06	1.805E+02	3.389E+00	9.454E+02	6.352E-02

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2074	4.677E+03	2.555E+06	1.717E+02	3.224E+00	8.993E+02	6.043E-02
2075	4.449E+03	2.430E+06	1.633E+02	3.066E+00	8.555E+02	5.748E-02
2076	4.232E+03	2.312E+06	1.553E+02	2.917E+00	8.138E+02	5.468E-02
2077	4.025E+03	2.199E+06	1.478E+02	2.775E+00	7.741E+02	5.201E-02
2078	3.829E+03	2.092E+06	1.405E+02	2.639E+00	7.363E+02	4.947E-02
2079	3.642E+03	1.990E+06	1.337E+02	2.511E+00	7.004E+02	4.706E-02
2080	3.465E+03	1.893E+06	1.272E+02	2.388E+00	6.662E+02	4.476E-02
2081	3.296E+03	1.800E+06	1.210E+02	2.272E+00	6.338E+02	4.258E-02
2082	3.135E+03	1.713E+06	1.151E+02	2.161E+00	6.028E+02	4.050E-02
2083	2.982E+03	1.629E+06	1.095E+02	2.055E+00	5.734E+02	3.853E-02
2084	2.837E+03	1.550E+06	1.041E+02	1.955E+00	5.455E+02	3.665E-02
2085	2.698E+03	1.474E+06	9.904E+01	1.860E+00	5.189E+02	3.486E-02
2086	2.567E+03	1.402E+06	9.421E+01	1.769E+00	4.936E+02	3.316E-02
2087	2.442E+03	1.334E+06	8.962E+01	1.683E+00	4.695E+02	3.155E-02
2088	2.322E+03	1.269E+06	8.525E+01	1.601E+00	4.466E+02	3.001E-02
2089	2.209E+03	1.207E+06	8.109E+01	1.523E+00	4.248E+02	2.854E-02
2090	2.101E+03	1.148E+06	7.713E+01	1.448E+00	4.041E+02	2.715E-02
2091	1.999E+03	1.092E+06	7.337E+01	1.378E+00	3.844E+02	2.583E-02
2092	1.901E+03	1.039E+06	6.979E+01	1.311E+00	3.656E+02	2.457E-02
2093	1.809E+03	9.881E+05	6.639E+01	1.247E+00	3.478E+02	2.337E-02
2094	1.720E+03	9.399E+05	6.315E+01	1.186E+00	3.308E+02	2.223E-02
2095	1.637E+03	8.941E+05	6.007E+01	1.128E+00	3.147E+02	2.115E-02
2096	1.557E+03	8.505E+05	5.714E+01	1.073E+00	2.994E+02	2.011E-02
2097	1.481E+03	8.090E+05	5.436E+01	1.021E+00	2.848E+02	1.913E-02
2098	1.409E+03	7.695E+05	5.170E+01	9.709E-01	2.709E+02	1.820E-02
2099	1.340E+03	7.320E+05	4.918E+01	9.236E-01	2.577E+02	1.731E-02
2100	1.275E+03	6.963E+05	4.678E+01	8.785E-01	2.451E+02	1.647E-02
2101	1.212E+03	6.623E+05	4.450E+01	8.357E-01	2.331E+02	1.566E-02
2102	1.153E+03	6.300E+05	4.233E+01	7.949E-01	2.218E+02	1.490E-02
2103	1.097E+03	5.993E+05	4.027E+01	7.562E-01	2.110E+02	1.417E-02
2104	1.044E+03	5.701E+05	3.830E+01	7.193E-01	2.007E+02	1.348E-02
2105	9.926E+02	5.423E+05	3.644E+01	6.842E-01	1.909E+02	1.283E-02
2106	9.442E+02	5.158E+05	3.466E+01	6.508E-01	1.816E+02	1.220E-02
2107	8.982E+02	4.907E+05	3.297E+01	6.191E-01	1.727E+02	1.160E-02
2108	8.544E+02	4.667E+05	3.136E+01	5.889E-01	1.643E+02	1.104E-02
2109	8.127E+02	4.440E+05	2.983E+01	5.602E-01	1.563E+02	1.050E-02
2110	7.731E+02	4.223E+05	2.838E+01	5.329E-01	1.487E+02	9.988E-03
2111	7.354E+02	4.017E+05	2.699E+01	5.069E-01	1.414E+02	9.501E-03
2112	6.995E+02	3.821E+05	2.568E+01	4.822E-01	1.345E+02	9.038E-03
2113	6.654E+02	3.635E+05	2.442E+01	4.586E-01	1.280E+02	8.597E-03